ENTERPRISE PLATFORM SPEC

HYPN CMMS & INDUSTRIAL IoT

Smart Operations, Asset Lifecycle & SCADA System

A comprehensive technical and functional architectural manual detailing the full-stack 4M Asset & Facility Management platform, real-time SCADA digital twins, IoT telemetry ingestion, work order lifecycle, and multi-tier granular access control system.

PROJECT NAME **HYPN / GMark-IoT Platform**	DOCUMENT TYPE **Complete Technical Manual**	VERSION **v2.6 Enterprise Edition**
DATE **October 2026**	FRAMEWORK **NestJS 10 + React 19 + Prisma**	TELEMETRY **MQTT + WebSockets + Modbus**

DOCUMENT OUTLINE & EXECUTIVE SECTIONS

Sec	Chapter / Topic	Summary of Contents
01	**Executive Summary & Platform Scope**	Core purpose, unified IoT & 4M operations, target capabilities
02	**System Architecture & Technology Stack**	Frontend, backend, real-time protocols, database, and telemetry pipeline
03	**4M Asset & Facility Management Modules**	Machines, multi-tier spatial facilities, spares, suppliers, and workforce
04	**Maintenance Lifecycle (Work Orders & Schedules)**	Preventative cron schedules, checkpoint masters, approval matrices, revisions
05	**Industrial SCADA, Digital Twins & IoT Telemetry**	6 pump systems, real-time WebSocket sync, alert thresholds, escalation policies
06	**Granular Role-Based Access Control (RBAC)**	5 system roles, module matrix, and hierarchical asset-level tree permissions
07	**Database Schema & Key Entities**	Prisma data model breakdown, entity relationships, and indexing strategy
08	**API Endpoints & Integration Reference**	REST controller matrix, WebSocket events, and telemetry payload formats

GMark Software Solutions © 2026. All Rights Reserved.

1. Executive Summary & Platform Scope

Bridging physical industrial hardware with intelligent computerized maintenance management.

The HYPN CMMS & Industrial IoT Platform (GMark-IoT) is a modern, enterprise-grade operating system designed for mission-critical infrastructure, commercial facilities, residential societies, and manufacturing plants. It provides an end-to-end bridge connecting physical sensors, smart controllers, and pumping infrastructure with sophisticated Computerized Maintenance Management System (CMMS) workflows.

By merging live telemetry ingestion with the 4M operational framework (Man, Machine, Material, Method), the platform empowers facility managers, chief engineers, and operators to prevent catastrophic equipment failures, automate maintenance routines, trace spare parts usage, and enforce security policies across complex multi-facility spatial hierarchies.

Core Philosophy: Unified Asset Intelligence

Traditional CMMS software operates in isolation from physical machine telemetry, while traditional SCADA systems lack business logic, work order routing, and inventory tracking. The HYPN platform combines both domains into a single pane of glass with sub-second WebSocket telemetry updates and automated condition-based ticketing.

2. System Architecture & Technology Stack

Full-stack distributed architecture built for resilience, scalability, and sub-second latency.

Layer	Technology	Role & Architectural Responsibility
Frontend Framework	**React 19 / TypeScript / Vite**	High-performance SPA with client-side routing, virtualized tables, and reactive UI
State & Comms	**Redux Toolkit + Socket.io Client**	Global session management, auth token handling, and live SCADA telemetry streaming
Styling & Visuals	**Tailwind CSS + Lucide + Three.js**	Bespoke design system, crisp iconography, and interactive 3D digital twins
Backend Engine	**NestJS 10 (TypeScript)**	Modular enterprise framework with dependency injection, guards, filters, and interceptors
Database & ORM	**PostgreSQL 15+ & Prisma ORM 5**	Relational database with strict type safety, migrations, and declarative relation mapping
IoT & Telemetry	**MQTT Client + Modbus Parser**	Ingests packet frames from industrial controllers, decodes telemetry registers, and broadcasts state
Authentication	**JWT + Refresh Rotation + bcrypt**	Stateless access tokens (15m), secure HTTP refresh cookies, and salted password hashing
Reporting & Export	**jsPDF, jspdf-autotable, SheetJS**	Real-time client-side generation of high-resolution PDF dossiers and Excel spreadsheets
API Documentation	**NestJS Swagger / OpenAPI 3**	Interactive API explorer accessible at /api/docs with bearer JWT authorization

GMark Software Solutions © 2026. All Rights Reserved.

3. 4M Asset & Facility Management Modules

Comprehensive computerized tracking across Machines, Facilities, Materials, and Personnel.

3.1 Machine Management (Asset Registry)

The Machine Management module serves as the central digital register for all industrial machinery, pumps, motors, and hardware assets. Each machine profile encompasses identification codes, operational statuses, parent-child dependencies, and technical parameters.

- **Auto-Code Generation:** Automated sequential codes (e.g. MCH001, MCH002) previewed live and locked at creation.
- **IoT Device Binding:** Binds machines to physical IoT MAC IDs to sync live telemetry, operating state, and alert logs.
- **Hierarchical Placement:** Allows machines to belong to specific facilities or function as parent-child modular sub-assemblies.
- **Tabular Customization:** Column visual customizer supporting 25+ parameters with column-wise instant filtering.
- **Technical Specs & Docs:** Attaches dynamic technical specifications (voltage, flow rate, RPM) and downloadable manuals/photos.

3.2 Facility Hierarchy (Spatial Architecture)

Assets exist within a structured, multi-tier spatial tree that accurately reflects real-world physical infrastructure.

Tier Level	Entity Type	Description & Relationships
Tier 1	**Main Facility (Campus / Plant)**	The top-level facility or plant campus (e.g., "Building 1", "North Manufacturing Complex").
Tier 2	**Sub-Facility (Building / Floor)**	Spatial partitions inside a Main Facility (e.g., "Floor 1", "Basement Pump Room"). Can be recursively nested.
Tier 3	**Child Sub-Facility**	Sub-sections within a facility (e.g., "HVAC Enclosure", "Booster Skid Bay").
Tier 4	**Machine / IoT Device**	Physical assets residing directly inside a sub-facility or directly under a main facility skid.

The facility module offers dual operational views: an Interactive Drag-and-Drop Tree (AssetTree) for spatial hierarchy visualization and re-parenting, alongside a dense Grid Table for batch audits and column searches.

3.3 Spares, Suppliers & Workforce Modules

- **Spare Parts & BOM:** Inventory registry tracking stock quantities, minimum safety thresholds, unit rates, part numbers, and machine-to-spare Bill of Materials linkages.
- **Supplier / Vendor Master:** Verified vendor directory tracking GST credentials, business categories, contact dossiers, and transaction logs.
- **Employee Master:** Personnel directory tracking employee codes, departments, designations, official emails, and manager reporting lines.

GMark Software Solutions © 2026. All Rights Reserved.

4. Maintenance Lifecycle (Work Orders & Schedules)

Rigorous preventive and corrective maintenance pipelines ensuring zero downtime.

4.1 Work Order Management & Multi-Tier Approvals

Work Orders (WOs) orchestrate all corrective and preventative maintenance tasks. The system implements a strict approval matrix to ensure work is verified by authorized supervisors and level-3 executives before execution and closure.

State / Step	Trigger / Actor	Operational Meaning & System Action
Draft / Created	**Technician or Automated Cron**	Ticket generated with machine ID, problem description, priority (Low/Medium/High/Critical), and estimated dates.
Pending Approval	**Supervisor Review**	Awaiting review by facility supervisor or maintenance manager.
Level-3 Approval	**Executive Approver**	For high-impact or high-budget maintenance tasks, requires Level-3 authorized sign-off.
Approved / Assigned	**Dispatcher / Supervisor**	Dispatched to assigned technician; notifications fired; start timer armed.
In Progress	**Field Technician**	Technician acknowledges task; logs spare parts used; records live observations.
Revision Requested	**Approver / QA Inspector**	Returned to technician for supplementary details, rework, or photo documentation.
Completed / Closed	**Supervisor / Inspector**	Verified completed; runtime logs committed; inventory counts deducted; archived.

4.2 Preventative Maintenance Schedules & Checkpoints Master

Preventative maintenance schedules automatically trigger Work Orders on recurring calendar frequencies (Daily, Weekly, Monthly, Quarterly, Semi-Annually, Annually). Each schedule incorporates:

- **Checkpoint Master Linkage:** Pre-configured checklists with standard inspection values, units of measurement, and tolerance limits (min/max thresholds).
- **Tolerance Window:** Pre-plan and post-plan tolerance day enforcement ensuring inspections occur within regulatory compliance windows.
- **Automatic Ticket Triggering:** A background cron engine continuously monitors upcoming schedules and dispatches work orders to assigned technicians.

GMark Software Solutions © 2026. All Rights Reserved.

5. Industrial SCADA, Digital Twins & IoT Telemetry

Real-time industrial plant supervision, 3D digital twins, and automated incident escalation.

5.1 Supported Industrial Plant Systems

The platform provides dedicated, pre-engineered SCADA dashboard templates for 6 critical building and plant services:

System Name	Key Monitored Registers	SCADA Visual Features
Domestic Water Booster	R/Y/B Voltage, Motor 1-4 Currents, Pressure (bar), Water Availability, VFD delay	Live 3D pump skid visualization, pump animation, pressure gauge, auto/manual mode
Fire Panel & Pumps	Line 1/2 Jockey & Main pump statuses, DG engine auto/manual, pressure transmitter	Fire alarm indicator, emergency trip annunciator, diesel engine battery voltage, run-hours
Sump Pit Pumping	High/Low water float switches, pump 1-2 fault types, motor run times, sump level	Pit liquid level meter, overflow hazard warning, alternate pump cycle sequencer
Subsoil Drainage	Percolation pit pressure, pump thermal overload, flow rate indicators	Drainage telemetry, duty-cycle balancer, pump runtime accumulator
Flushing Water System	Header pressure, booster pump status, suction tank level switch	Dual-pump lead-lag sequencing, dry-run protection status
Garden & Irrigation	Zone solenoid valve statuses, line pressure, soil moisture inputs	Zone scheduling matrix, automated pressure cutoff, water consumption tracking

5.2 Telemetry Ingestion Pipeline & Alarm Escalation Matrix

Physical IoT controllers publish payload packets over MQTT / Modbus. The backend telemetry processor decodes byte arrays, updates the DeviceState table, and pushes real-time WebSocket diffs to connected clients.

When an abnormal register threshold is encountered (e.g. Pump Overload, Input Power Failure, Dry Run), the Alert Matrix triggers a multi-tier escalation:

- **Tier 1 (Instant):** On-screen visual toast & audible alarm; notification sent to on-duty Field Operator.
- **Tier 2 (15 Minutes):** If unacknowledged, ticket auto-created; automated email alert dispatched to Building Administrator.
- **Tier 3 (60 Minutes):** High-priority SMS/Email notification escalated to Super Admin and Chief Operations Engineer.

GMark Software Solutions © 2026. All Rights Reserved.

6. Granular Role-Based Access Control (RBAC)

Multi-level security governance ensuring strict data segregation across multi-tenant assets.

6.1 Five System Roles & Access Baselines

Role Identifier	Scope of Authority	Default Baseline Capabilities
SUPER_ADMIN	**Platform-wide / Global**	Full unrestricted read/write/delete/configure access across all modules, tenants, and facilities.
SOCIETY_ADMIN	**Plant / Society Enterprise**	Administrative access to assigned facilities, users, machines, and configuration masters.
BUILDING_ADMIN	**Specific Building / Facility**	Operational control over assigned building infrastructure, work order approval, and machines.
OPERATOR	**Field Execution / Plant Shift**	Can view plant telemetry, acknowledge SCADA alarms, update assigned work orders, and log check points.
VIEWER	**Auditor / Read-Only Client**	Strictly view-only access to assigned dashboards, machines, and maintenance histories.

6.2 Multi-Dimensional Permission Matrix & Granular Asset Trees

The platform supports three distinct layers of permission resolution:

- **1. Module & Action Matrix:** Governs 7 distinct actions across every module and sub-tab: View, Create, Edit, Delete, Import, Export, Print.
- **2. Designation & User Overrides:** Designations provide role blueprints; individual UserPermission records allow precise exceptions for individual user accounts.
- **3. Hierarchical Asset-Level Permissions:** Restricts access down to individual physical nodes (Main Facility, Sub-facility, Machine).

Granular Asset Permission Hierarchy Engine (hasAssetPermission)

The asset permission engine evaluates permissions recursively from child to parent:
1. Explicit Machine Check: asset_mac_${id} or asset_dev_${deviceId} -> if explicit canView is defined, it takes immediate precedence.
2. Parent Facility Check: If machine has no explicit rule, inherits from asset_fac_${facilityId}.
3. Parent Main Facility Check: If facility has no explicit rule, inherits from asset_mf_${mainFacilityId}.
4. Global Asset Root Check: Inherits from root "assets" permission if defined; otherwise defaults to denied.
This guarantees that users granted access to only a specific machine or facility cannot see or tamper with sibling assets.

GMark Software Solutions © 2026. All Rights Reserved.

7. Database Schema & Key Entities

Prisma ORM data model relationships powering the entire ecosystem.

Model Entity	Key Fields & Types	Relations & Purpose
User	id, email, password, role, designation, facilityId, deviceId	Primary auth identity. Relations: UserPermissions, WorkOrders, UserBuilding.
Employee	id, employeeCode, fullName, designation, department, adminId	Staff directory. Relations: Users, WorkOrders, AssignedMachines.
MainFacility	id, mainFacilityCode, mainFacilityName, category, status	Top-level plant/campus. Relations: Facilities, Buildings, Machines, Users.
Facility	id, facilityCode, facilityName, mainFacilityId, parentFacilityId	Spatial sub-facility/floor. Recursive parent/child. Relations: Machines, Users.
Machine	id, machineCode, name, deviceId, facilityId, parentMachineId	Asset registry. Relations: Facility, Device, Documents, Specs, WorkOrders.
Device	id, deviceId (MAC), name, latestData, lastSeen	Physical IoT hardware. Relations: DeviceState, Alerts, AlertConfigs, Values.
DeviceState	id, deviceId, motor1-4Status, inputPowerFault, pressureBar	Real-time telemetry cache for ultra-fast SCADA dashboard rendering.
WorkOrder	id, code, problemDescription, priority, status, assignedTechId	Maintenance task lifecycle. Relations: Creator, Tech, Approvers, Revisions.
MaintenanceSchedule	id, scheduleCode, frequency, scheduleType, machineId	Recurring preventative cron jobs. Relations: Tasks, Logs, Machine.
CheckpointMaster	id, code, description, frequency, standardValue, min/max	Checklist items library. Relations: ScheduleTasks, UserCheckpoints.
SparePart	id, partNumber, partName, stockQuantity, unitCost, supplierId	Inventory registry. Relations: Supplier, MachineSpares.
Supplier	id, supplierCode, name, gstNumber, contactPerson, email	Vendor directory. Relations: SpareParts, PurchaseOrders.
Designation	id, name, isActive, adminId	Role blueprint. Relations: Permissions (moduleId, canView, canEdit...).
UserPermission	id, userId, moduleId, canView, canCreate, canEdit, canDelete	Per-user override matrix for modules, sub-tabs, and asset nodes.
AuditLog	id, userId, action, module, ipAddress, createdAt	Immutable compliance audit trail tracking user interactions and mutations.

GMark Software Solutions © 2026. All Rights Reserved.

8. API Endpoints & Deployment Guide

REST interface reference and production operations handbook.

8.1 RESTful Endpoint Matrix

HTTP Verb & Route	Guards & Auth	Functionality & Response
POST /auth/login	Public	Validates credentials; issues JWT access token + refresh cookie; returns user profile + resolved permissions.
GET /auth/me	JwtAuthGuard	Validates session and returns refreshed user permissions matrix.
GET /machines	JwtAuthGuard, RolesGuard	Returns machines filtered by tenant adminId and granular asset permissions.
GET /machines/:id	JwtAuthGuard, RolesGuard	Returns single machine details, online telemetry state, specs, and docs.
GET /facilities	JwtAuthGuard	Returns hierarchical facilities list filtered by user asset permissions.
GET /main-facilities	JwtAuthGuard	Returns top-level campus facilities filtered by asset permissions.
GET /work-orders	JwtAuthGuard	Retrieves work orders filtered by status, priority, technician, or machine.
POST /work-orders	JwtAuthGuard	Creates maintenance work order; sends real-time dispatch notification.
GET /schedules	JwtAuthGuard	Retrieves preventative maintenance recurring schedules and checkpoint tasks.
GET /spares	JwtAuthGuard	Lists spare parts inventory, warehouse quantities, and supplier associations.
GET /designations	JwtAuthGuard, RolesGuard	Manages roles, permission blueprints, and assigned employee lists.
WS /scada (Socket.io)	JWT Handshake	Real-time telemetry streaming: emits "telemetry:update" and "alert:triggered".

8.2 Deployment & Environment Execution

To run the platform locally or in a staging environment:

- **Backend Startup:** cd d:\gmark-iot\backend && npm run start:dev (runs NestJS API on http://localhost:3000)
- **Frontend Startup:** cd d:\gmark-iot\frontend && npm run dev (runs Vite dev server on http://localhost:5173)
- **Database Migration:** npx prisma migrate dev && npx prisma db seed (seeds superadmin, roles, and sequences)
- **Production Build:** npm run build in backend (dist/main.js) and frontend (dist/ bundle for Nginx/CDN)

GMark Software Solutions © 2026. All Rights Reserved.